

PROBLEM SCENARIO

ITEC624 Network Security & Applications

Semester 2, 2025

Part 1: Skill based assessment (30 marks)

In this part, you need to find solutions for a number of practical questions. To solve the problems, you may use any security tools such as Bourne/Bash Shell, Wireshark, John the ripper, nmap, IP scanner etc.

Q1. Write a program in shell script that will take an integer value from the user as input and check whether the number is odd or even. The program should continue for n times. See below the sample input and output of the program. **[5 marks]**

```
How many times you want to enter the number? 3
Enter Number 1: 10
It is an even number!
Enter Number 2: 7
It is an odd number!
Enter Number 3: 0
Please enter a number greater than 0
```

Q2. Advanced Directory Permissions and Access Control using Kali Linux (Please review Lab 5 for your reference). [10 marks]

Scenario:

You are working as a cybersecurity consultant at CyberDefense Solutions. Your client organization is setting up a new penetration testing division with multiple specialized teams. Each team requires secure, isolated access to their respective testing environments and tools, while maintaining strict access controls to prevent unauthorized access to sensitive penetration testing data.

Tasks:

1. *Setting up the Environment: 1 mark*

- Create a main directory named PenTestLabs.
- Under PenTestLabs, create two sub-directories named NetworkSec and WebAppSec.

2. *Permission Setup for PenTestLabs: 1 mark*

- Ensure that the PenTestLabs directory has permissions rwxrwx---.
- Verify the permissions using `ls -ld /PenTestLabs/`.

3. User and Group Creation: 1 mark

- Create two users: pentester1 and pentester2.
- Create a group named netsec_team and add pentester1 to it.
- Create another group named webapp_team and add pentester2 to it.

4. Setting Up Permissions for Sub-directories: 2 marks

- **For NetworkSec:**
 - Ensure that pentester1 has full permissions.
 - Group netsec_team should have read, write, and execute permissions.
 - Others should have no permissions.
- **For WebAppSec:**
 - Ensure that pentester2 has full permissions.
 - Group webapp_team should only have read and execute permissions.
 - Others should have no permissions.

5. Permission Verification: 2 marks

- Use the `getfacl` command to verify permissions for both NetworkSec and WebAppSec.
- Also, check the permissions after logging in as pentester1 and pentester2 to ensure they can access their respective directories but not the other's directory.

6. Creating Shared Resources Directory: 2 marks

- Create a new directory called SharedTools under PenTestLabs.
- Set permissions so that both pentester1 and pentester2 can read and execute files, but only members of netsec_team can write to it.
- Verify that pentester2 can access files in SharedTools but cannot create new files there.

7. Implementing Supervisor Access: 1 mark

- Create a new user called pentest_supervisor.
- Configure permissions so that pentest_supervisor has read-only access to all directories (NetworkSec, WebAppSec, and SharedTools) without being able to modify any files.
- Test the access by logging in as pentest_supervisor and attempting to read files and create new files in each directory.

Deliverables:

- A detailed documentation of each step you took, including screenshots for verification.
- A brief reflection on how proper directory permissions contribute to maintaining confidentiality and integrity in penetration testing environments.
- *Additional:* Commentary on how the shared resources and supervisor access models could be applied in real-world penetration testing team structures.

Hints:

- Refer to the provided tutorial for commands and their syntax.

- Remember to switch to the root user when necessary to change permissions or add users.
- For the shared directory, consider using Access Control Lists (ACLs) with setfacl if standard permissions aren't sufficient.

Q3. Answer the following questions on password cracking. [5 marks]

- a) Try to guess the password of Act1.pdf file from the following conversation. [1 marks]

David: Hello, my name is David. It's nice to meet you.

Jenny: Hi, I'm Jenny. Nice to meet you too.

David: Jenny, nice name! Jenny, what do you do for a living?

Jenny: I work at the local school teaching English. What do you do for a living?

David: I'm also an English teacher. I teach at school.

Jenny: Wow! That's great!

David: You are looking pretty with those purple glasses.

Jenny: Thank you.

David: I guess purple is your favourite colour, right?

Jenny: That's right. My bus is coming. It has been really nice talking to you.

David: Thank you! Enjoy rest of your day!

Jenny: You too! Bye!

This is a conversation between Jenny and David at the bus stop. Use the information provided above to find the password of the Act1.pdf file. What is the name of the procedure that you followed to find the password in this activity?

- b) Download a pdf password cracker tool and retrieve the password of the Act2.pdf file. What is the password? What secret message did you retrieve from the file?

Hint: The length of the password is 6, and it uses only digits. [4 marks]

Q4. Advanced Network Reconnaissance and Vulnerability Assessment using Nmap [10 marks]

Scenario:

You have been hired as a junior penetration tester at SecureNet Auditing Inc. Your team is conducting a comprehensive security assessment of a client's internal network infrastructure. Your supervisor has assigned you to perform network reconnaissance using Nmap to identify potential security weaknesses before proceeding with deeper vulnerability testing.

Prerequisites:

- Ensure Nmap is installed on your Kali Linux system (it comes pre-installed).

- You will be scanning your own local network or a designated lab environment (e.g., Metasploitable VM).
- Verify: `nmap --version` to confirm installation.

Tasks:

1. Network Discovery and Host Enumeration (2 marks):

- Use Nmap to perform a ping sweep of your local network subnet to identify all active hosts.
- Document the total number of active hosts discovered.
- Provide a screenshot showing the IP addresses and response times of discovered hosts.
- **Hint:** Use `nmap -sn <IP range>` for host discovery without port scanning.

2. Service Version Detection (2 marks):

- Select one active host from Task 1 and perform a service version scan on the most common 1000 ports.
- Identify at least three services running on the target, including their version numbers.
- Provide a screenshot displaying the service names, versions, and associated ports.
- **Hint:** Use `nmap -sV <target IP>` for version detection.

3. Operating System Fingerprinting (2 marks):

- Conduct an OS detection scan on the same target host from Task 2.
- Document the operating system identified, including the OS family and any version details.
- Provide a screenshot showing the OS detection results.
- **Hint:** Use `nmap -O <target IP>` for OS detection.

4. Advanced Port Scanning Techniques (2 marks):

- Perform a SYN stealth scan (half-open scan) on the target host.
- Compare the results with a standard TCP connect scan on the same host.
- Document any differences in the results and explain why SYN scanning is considered "stealthier."
- Provide screenshots for both scan types.
- **Hint:** Use `nmap -sS <target IP>` for SYN scan and `nmap -sT <target IP>` for TCP connect scan.

5. Vulnerability Assessment with NSE Scripts (3 marks):

- Use Nmap Scripting Engine (NSE) to run vulnerability detection scripts against your target host.
- Execute the default script scan and identify at least two potential vulnerabilities or security concerns.
- Document the specific vulnerabilities found, affected services, and potential security implications.
- Provide a screenshot showing the NSE script output.
- **Hint:** Use `nmap -sV --script=default <target IP>` or `nmap -sC -sV <target IP>`.

Deliverables:

- A comprehensive documentation of each task with detailed screenshots showing command syntax and results.
- A brief reflection (1-2 paragraphs) on the importance of reconnaissance in penetration testing and how both attackers and defenders could use the information gathered.

Hints:

- Take clear, full-screen screenshots that show both the command used and complete results.

- Use sudo or root privileges for certain scan types (SYN scan, OS detection).
- If scanning takes too long, consider using faster timing templates like -T4 (e.g., nmap -T4 -sV <target IP>).

Part 2: Case Study (20 marks)

In this part, you have to write a network security analysis and planning report (1500 words) for a small organization. Please go through the [NIST report](#) (also uploaded on Canvas) before starting the assignment. You can choose any of the following organisation (but not limited to):

- ✓ A grocery shop
- ✓ A restaurant
- ✓ An art gallery
- ✓ An auto repair service
- ✓ An e-commerce portal
- ✓ A boutique house
- ✓ A day care service
- ✓ An interior farm
- ✓ A car resale company
- ✓ A party planner

Structure

Abstract: An abstract (a short summary of the article) needs to convey a complete synopsis of the paper, but within a word tight limit. Writing an abstract includes brief introduction to the general topic of the work and then explanation of the exact research question, including the aims. It should then include a description of the methodology, results, and discussion.

Introduction: In the Introduction, you are attempting to inform the reader about the rationale behind the work, justifying why your work is an essential component of research in the field. The introduction does not have a strict word limit, unlike the abstract, but it should be as concise as possible. It can be a tricky part of the paper to write, so many scientists and researchers prefer to write it last, ensuring that they miss no major points. The introduction gives an overall review of the paper but does address a few slightly different issues from the abstract. An introduction should emphasize on background, importance, limitations, and assumptions. *You should provide a short overview of the selected business organisation in this section.*

Understanding and Managing Risks: For this portion of your report you should identify elements of risks associated with the business. You also need to provide detail description of how you will be managing the risks and discuss impacts of threats and vulnerabilities on the business.

Protecting Business Resources: This section will provide a number of actions that will be taken to safeguard business information and resources. This may include identify access to business information, create policies and procedures, implement security measures to protect business information and resources, detect intrusion, respond

and recovery strategies to internal and external threats. *You need to relate data stewardship with protection of business resources to address GA2.*

Creating a safe and secure work environment: This section will provide a guideline for users to keep work environment safe from cyber-attacks. It will also suggest the actions to be taken if any suspicious behavior is noticed by a user in the organisation's computer or network systems.

Conclusion: This is really just a more elaborate version of the abstract. In a few paragraphs you should summarize your findings and recommendations. Your abstract will do most of this for you but, as long as you do not get carried away, especially for longer reports, it can help the reader absorb your findings a little more.

References: Please refer to the referencing section in the AT3 assessment guide.